JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON THURSDAY 02nd APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

CHEMISTRY — TEST PAPER WITH SOLUTION

SECTION-A

51. The mass of iron converted into Fe_3O_4 by the action of 18 g of steam is :

(Given : Molar mass of H, O and Fe are 1, 16 and 56 g mol^{-1} respectively)

Assume iron is present in excess :

(1) 2.1 g (2) 4.2 g

(3) 21 g (4) 42 g

Sol. $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

$n_{steam} = \frac{18}{18} = 1$

n_{Fe} required $= \frac{3}{4}$

mass of Fe required $= \frac{3}{4} \times 56$

$= 42$ gm

52. What is the energy (in J $atom^{-1}$) required for the following process ?

$Li^{2+}(g) \rightarrow Li^{3+}(g) + e^-$

(Take the ionization energy for the H atom in the ground state as 2.18×10^{-18} J $atom^{-1}$)

(1) 8.72×10^{-18}

(2) 1.962×10^{-18}

(3) 1.962×10^{-17}

(4) 6.54×10^{-17}

Sol. $E_{Li^{2+}} = E_H \times z^2$

$= 2.18 \times 10^{-18} \times (3)^2$

$= 1.962 \times 10^{-17}$ J $atom^{-1}$

53. Given below are two statements :

Statement (I) : The correct sequence of bond lengths in the following species is :

$O_2^+ < O_2 < O_2^- < O_2^{2-}$

Statement (II) : The correct sequence of number of unpaired electrons in the following species is :

$O_2 > O_2^+ > O_2^- > O_2^{2-}$

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol.

Species	Bond order	unpaired electrons
$O_2^{\oplus}$	2.5	1
O_2	2	2
O_2^-	1.5	1
O_2^{2-}	1	0

54. Consider the following data

(i) $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g) +$ 1200 kJ/mol.

(ii) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g) + 164$ kJ/mol.

(iii) $HCl(g) + aq \rightarrow HCl(aq) + 83$ kJ/mol.

(iv) $Al_2Cl_6(s) + aq \rightarrow Al_2Cl_6(aq) + 663$ kJ/mol.

The enthalpy of formation of anhydrous solid Al_2Cl_6 is :

(1) –648 kJ mol^{-1} (2) –1350 kJ mol^{-1}

(3) –2002 kJ mol^{-1} (4) –1527 kJ mol^{-1}

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Sol. $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g)$

$\Delta_rH_1 = -1200$ kJ/mol

$[H_2(g) + Cl_2(g) \rightarrow 2HCl(g)] \times 3$

$\Delta_rH_2 = -163 \times 3$ kJ/mol

$[HCl(g) + aq \rightarrow HCl(aq)] \times 6$

$\Delta_rH_3 = -83 \times 6$ kJ/mol

$Al_2Cl_6(aq) \rightarrow Al_2Cl_6(s) + aq.\ \Delta_rH_4 = +663$ kJ/mole

$2Al(s) + 3Cl_2 \rightarrow Al_2Cl_6(s) \quad \Delta_fH$

$\Delta_fH = \Delta_rH_1 + \Delta_rH_2 + \Delta_rH_3 + \Delta_rH_4$

$= -1200 - 163 \times 3 - 83 \times 6 + 663$

$= -1527$ kJ/mole

55. 19.5 g of fluoro acetic acid (molar mass = 78 g mol^{-1}) is dissolved in 500g of water at 298 K. The depression in the freezing point of water was 1°C. What is K_a of fluoro acetic acid ?

(For water, $K_f = 1.86$ K kg mol^{-1}). Assume molarity and molality to have same values.

(1) 10^{-6} (2) 4×10^{-4}

(3) 3×10^{-5} (4) 3×10^{-3}

Sol. $\Delta T_f = iK_fm$

$$\text{no. of moles} = \frac{19.5}{78} = \frac{1}{4}\text{mole}$$

$$1 = i \times 1.86 \times \frac{1/4}{1/2}$$

$$i = \frac{2}{1.86}$$

$i = 1 + (n-1)\alpha$

$i = 1 + \alpha$

$$\alpha = \frac{2}{1.86} - 1 = 0.075$$

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

$$= \frac{1}{2} \times \frac{(0.075)^2}{(1-0.075)}$$

$= 3 \times 10^{-3}$

56. The solubility product constants of Ag_2CrO_4 and AgBr are 32x and 4y respectively at 298 K.

The value of $\left(\frac{\text{molarity of } Ag_2CrO_4}{\text{molarity of AgBr}}\right)$ can be expressed as :

(1) $\frac{2\sqrt[3]{x}}{y}$ (2) $2\sqrt{\frac{x}{y}}$

(3) $\sqrt{\frac{x}{y}}$ (4) $\frac{\sqrt[3]{x}}{\sqrt{y}}$

Sol. For $Ag_2C_2O_4$

$4S_1^3 = 32x$

$\Rightarrow S_1 = 2x^{1/3}$

For AgBr

$S_2^2 = 4y$

$\Rightarrow S_2 = 2y^{1/2}$

Ratio $\left(\frac{S_1}{S_2}\right) = \frac{\sqrt[3]{x}}{\sqrt{y}}$

57. An electrochemical cell is constructed using half cells in the direction of spontaneous change

$Fe(OH)_2(s) + 2e^- \rightarrow Fe(s) + 2OH^-(aq)\ \ E^0 = -0.88V$

and $AgBr(s) + e^- \rightarrow Ag(s) + Br^-(aq)\ \ E^0 = +0.07V$

Which of the following option is **correct** ?

(1) Overall reaction $Fe(s) + 2OH^-(aq) + 2AgBr(s) \rightleftharpoons Fe(OH)_2(s) + 2Ag(s) + 2Br^-(aq)$

(2) $E^0_{cell} = -0.95V$

(3) Fe is reduced in the electrochemical cell

(4) E^0_{cell} is an extensive property

Sol. Overall reaction

$Fe(s) + 2OH^-(aq) + 2AgBr(s) \rightleftharpoons Fe(OH)_2(s) + 2Ag(s) + 2Br^-(aq)$

$E^o_{cell} = E^o_{R(cathode)} - E^o_{R(anode)}$

$= 0.07 - (-0.88) = 0.95$ V

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58. $t_{100\%}$ is the time required for the 100% completion of the reaction while $t_{1/2}$ is the time required for 50% of the reaction to be completed. Which of the following option correctly represents the relation between $t_{100\%}$ and $t_{1/2}$ for zero and first order reactions respectively ?

(1) $t_{100\%} = (t_{1/2})^2$ and $t_{100\%} = (t_{1/2})^{-\infty}$

(2) $t_{100\%} = 2t_{1/2}$ and $t_{100\%} = (t_{1/2})^{\infty}$

(3) $t_{100\%} = 2t_{1/2}$ and $t_{100\%} = (2t_{1/2})^2$

(4) $t_{100\%} = (t_{1/2})^{\infty}$ and $t_{100\%} = 2t_{1/2}$

Sol. **Zero –order**

$$t_{100\%} = \frac{a}{k};\ t_{1/2} = \frac{a}{2k}$$

$$t_{100\%} = 2\,t_{1/2}$$

Ist order

$t_{100\%} = \infty$

$$t_{1/2} = \frac{\ell n 2}{k}$$

$t_{100\%} = (t_{1/2})^{\infty}$

59. Given below are two statements :

Statement (I) : The first ionisation enthalpy of the elements Na, Mg, Cl and Ar follows the order Na > Mg > Cl > Ar

Statement (II) : Among Ca, Al, Fe and B, the third ionisation enthalpy is very high for Ca.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. (i) Na < Mg < Cl < Ar → I^{st} ionization energy

(ii) Among B, Ca, Al, Fe

Ca has maximum 3rd ionization energy among given due to inert gas configuration.

60. Given below are two statements :

Statement (I) : Oxidising power of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$, which is the basis of "Layer test".

Statement (II) : "Layer test" to identify Br_2 and I_2 in aqueous solution involves the oxidation of bromide or iodide into Br_2 or I_2 respectively with Cl_2, which is a type of displacement redox reaction.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. (I) $F_2 > Cl_2 > Br_2 > I_2$ (Oxidizing power order)

(II) $NaBr + Cl_2\text{-water} \rightarrow Br_2 + NaCl$

$NaI + Cl_2\text{-water} \rightarrow I_2 + NaCl$

61. Which of the following sets includes all the species that will change the orange colour of $K_2Cr_2O_7$ in acidic medium ?

(1) Fe^{2+}, Sn^{2+}, I^-, S^{2-}

(2) S^{2-}, Fe^{3+}, I^-, $C_2O_4^{2-}$

(3) Fe^{2+}, NO_2^-, SO_2, Sn^{4+}

(4) Fe^{3+}, SO_4^{2-}, S^{2-}, Sn^{4+}

Sol. $\text{Reducing agent} + KmnO_4 \xrightarrow[\text{medium}]{\text{acidic}} Mn^{+2} + \text{Product}$

Given Reducing Agent

$Fe^{+2} \rightarrow Fe^{+3}$

$Sn^{+2} \rightarrow Sn^{+4}$

$I^- \rightarrow I_2$

$S^{-2} \rightarrow S$

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62. Match **List – I** with **List – II**.

List – I **Chromium (III) Complexes (en = ethylene diamine)**		**List – II** $\Delta_0(cm^{-1})$	
A.	$[Cr(CN)_6]^{3-}$	I.	15,060
B.	$[CrF_6]^{3-}$	II.	17,400
C.	$[Cr(H_2O)_6)]^{3+}$	III.	22,300
D.	$[Cr(en)_3]^{3+}$	IV.	26,600

Choose the **correct** answer from the options given below :

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-I, C-II, D-III

Sol. CFSE is proportional to strength of ligand followed by chelation.

CN^- is stronger ligand than en

$CN^- > en > H_2O > F^-$

63. Given below are two statements :

Statement (I) : 1,2,3-Trihydroxypropane can be separated from water by simple distillation.

Statement (II) : An azotropic mixture cannot be separated by fractional distillation.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. Glycerol is separated by distillation under reduced pressure.

64. Given below are two statements :

Statement (I) : Benzyl chloride reacts faster in S_N1 mechanism than ethyl chloride.

Statement (II) : Ethyl carbocation intermediate is less stabilized by hyperconjugation than benzyl carbocation by resonance.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. Rate of $SN^1 \propto$ Stability of carbocation

Order of stability of carbocation :

$Ph-\overset{\oplus}{C}H_2 > CH_3-\overset{\oplus}{C}H_2$

Hyperconjugation in $CH_3-\overset{\oplus}{C}H_2$

Resonance in $Ph-\overset{\oplus}{C}H_2$

65. In IUPAC nomenclature, the **correct** order of decreasing priority of functional group is :

(1) $-CONH_2$, $>C=O$, $-CHO$, $-NH_2$, $-C\equiv C-$

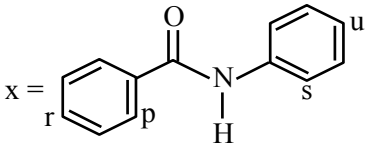
(2) $-CONH_2$, $-COOCH_3$, $-CHO$, $-NH_2$, $-OH$

(3) $-CONH_2$, $-CHO$, $>C=O$, $-NH_2$, $-C\equiv C-$

(4) $-CONH_2$, $-CHO$, $-CN$, $-NH_2$, $-C\equiv C-$

Sol. Theory based

66. For the given molecule, "x", the preferred site for the attack of the electrophile is :



(1) Predominantly at "r"

(2) "r" and "u"

(3) "p" and "s"

(4) Predominantly at "u"

Sol. U is predominant site towards attack of electrophile because of +M nature of nitrogen.

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67. Match **List-I** with **List-II**.

	List-I Mixture of Compounds		**List-II Reagent used to distinguish**
A.	Diethyl amine + Ethyl amine	I.	Bromine water
B.	Acetaldehyde + Acetone	II.	$CHCl_3$ + KOH, Δ
C.	Ethanol + Phenol	III.	Neutral $FeCl_3$
D.	Benzoic acid + Cinnamic acid	IV.	Ammonical silver nitrate

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-I, D-III

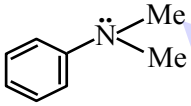
(2) A-IV, B-II, C-III, D-I

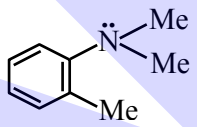
(3) A-II, B-IV, C-I, D-III

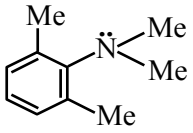
(4) A-II, B-IV, C-III, D-I

Sol.
- Phenol gives violet colour with neutral $FeCl_3$ not ethanol.
- 1° amine gives positive test with $CHCl_3$/KOH not 2° amine.
- Aldehyde gives positive test with ammonical silver nitrate not ketone.
- Cinnamic acid gives bromine water test.

68. Consider the three aromatic molecules (P, Q and R) whose structures have been given below :

P. 

Q. 

R. 

The **correct** order regarding the reactivity of these compounds with $Ph–\overset{(+)}{N}\equiv N\ Cl^{(-)}$ under optimum but slightly acidic medium is :

(1) P > Q > R (2) R > P > Q

(3) R > Q > P (4) P > R > Q

Sol. Rate of ESR ∝ nucleophilicity of benzene ring.

69. Match **List-I** with **List-II**.

	List-I Vitamin		**List-II Name**
A.	Vitamin B_1	I.	Pyridoxine
B.	Vitamin B_2	II.	Ascorbic acid
C.	Vitamin B_6	III.	Thiamine
D.	Vitamin C	IV.	Riboflavin

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-I, B-III, C-II, D-IV

Sol. Theory based

70. A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and treated with K_2CrO_4 to give yellow precipitate. When the sodium carbonate extract of the salt solution is heated with conc. HNO_3 and ammonium molybdate, it resulted a canary yellow precipitate. The cation and anion present in the salt are respectively.

(1) Ca^{2+} and SO_4^{2-} (2) Ba^{2+} and PO_4^{3-}

(3) Mn^{2+} and PO_4^{3-} (4) Ba^{2+} and SO_4^{2-}

Sol. (i) Ba^{+2} gives apple green colour while Ca^{+2} and Sr^{+2} gives brick red and crimson red in flame test.

(ii) $Ba^{+2} + CrO_4^{2-} \rightarrow \underset{\text{Yellow ppt}}{BaCrO_4}$

(iii) $PO_4^{3-} + \text{Conc } HNO_3 + (NH_4)_2MoO_4 \longrightarrow \underset{\text{Canery yellow ppt}}{(NH_4)_3[P(Mo_3O_{10})_4]}$

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SECTION-B

71. 5.33 g of $CrCl_3 \cdot 6H_2O$, which is a 1 : 3 electrolyte, is dissolved in water and is passed through a cation exchanger. The chloride ions in the eluted solution, on treatment with $AgNO_3$ results in 8.61 g of AgCl. The ratio of moles of complex reacted and moles of AgCl formed is _____ $\times 10^{-2}$. (Nearest integer)
[Molar mass in g mol^{-1} Cr : 52, Ag : 108, Cl : 35.5, H : 1, O : 16]

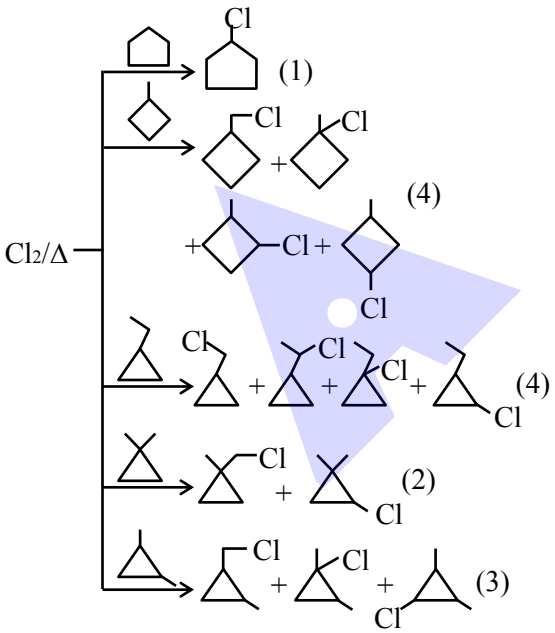
Sol. $[Cr(H_2O)_6]Cl_3 + \underset{\text{excess}}{AgNO_3} \rightarrow \underset{\text{ppt}}{3AgCl(s)}$

$$\frac{\text{no. of moles of complex reacted}}{\text{no. of moles of AgCl precipitated}} \times 100$$

$$= \frac{1}{3} \times 100 = 33.3$$

72. Consider the isomers of hydrocarbon with molecular formula C_5H_{10}. These isomers do not decolourise $KMnO_4$ solution. These isomers are subjected to chlorination with chlorine in presence of light to give monochloro compounds. The total number of monochloro compounds (structural isomers only) formed is _____.

Sol.



$1 + 4 + 4 + 2 + 3 = 14$

73. One mole of an alkane (x) requires 8 mole oxygen for complete combustion. Sum of number of carbon and hydrogen atoms in the alkane (x) is _____.

Sol. $C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \rightarrow nCO_2 + (n+1)H_2O$

$$1 \qquad \left(\frac{3n+1}{2}\right)$$

$$\frac{3n+1}{2} = 8 \Rightarrow 3n + 1 = 16$$

$$\Rightarrow 3n = 15$$

$$n = 5$$

So alkane is $C_5H_{12} \Rightarrow 17$

74. For reaction $A \rightarrow P$, rate constant $k = 1.5 \times 10^3$ s^{-1} at 27°C. If activation energy for the above reaction is 60 kJ mol^{-1}, then the temperature (in °C) at which rate constant, $k = 4.5 \times 10^3$ s^{-1} is _____. (Nearest integer)
Given : log 2 = 0.30, log 3 = 0.48, $R = 8.3$ J K^{-1} mol^{-1}, ln 10 = 2.3

Sol. $$\ell n\left(\frac{K_2}{K_1}\right) = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

$$\ell n\left(\frac{4.5\times10^3}{1.5\times10^3}\right) = \frac{60\times10^3}{8.3}\left[\frac{T_2 - 300}{300.T_2}\right]$$

$$\ell n3 = \frac{60\times10^3}{8.3}\left[\frac{T_2 - 300}{300.T_2}\right]$$

$\Rightarrow T_2 = 314.4$ K

$\Rightarrow t = 41.4$°C

75. At the transition temperature T, $A \rightleftharpoons B$ and $\Delta G^0 = 105 - 35 \log T$ where A and B are two states of substance X. The transition temperature in °C when pressure is 1 atm is _____. (Nearest integer)

Sol. $\Delta G° = 0$ (eqm. at standard pressure)

$105 - 35 \log T = 0$

$\log T = 3$

$T = 1000$ K

$T = 727$°C

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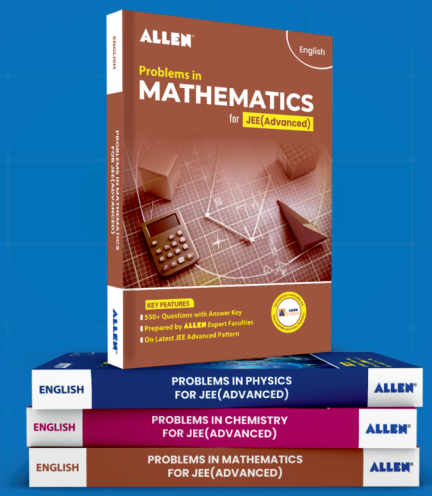


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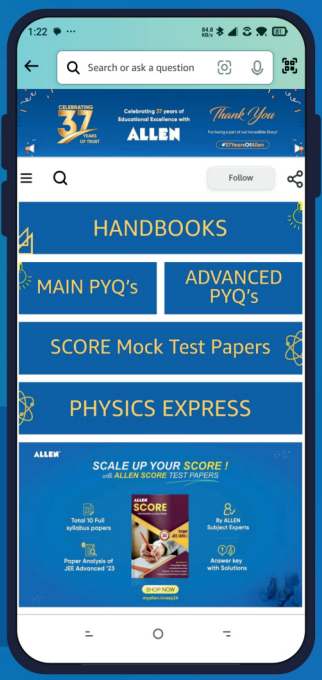
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